

Pearson Edexcel International Advanced Level

Monday 13 October 2025

Morning (Time: 1 hour 30 minutes)

Paper
reference

WCH12/01A

Chemistry

International Advanced Subsidiary/Advanced Level

**UNIT 2: Energetics, Group Chemistry,
Halogenoalkanes and Alcohols
Question Paper**

You must have:

Scientific calculator, Data Booklet, ruler and Answer Book

Information

- A Periodic Table is printed on the back cover of this paper.

Turn over ►

P87419A

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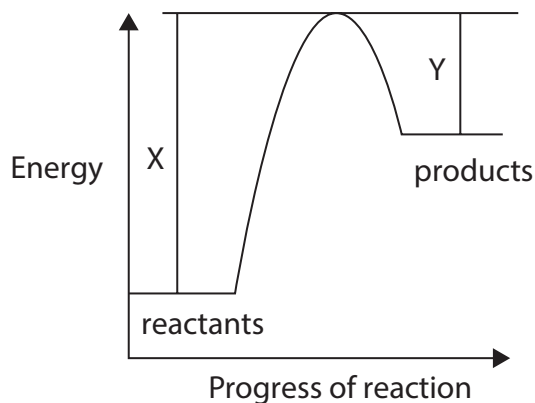



Pearson

SECTION A

Answer ALL the questions in this section in the Answer Booklet.

- 1 The reaction profile diagram for a reaction is shown.

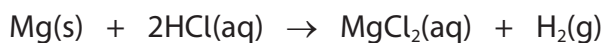


What is the minimum energy needed for this reaction to occur?

- A X
- B Y
- C $X - Y$
- D $X + Y$

(Total for Question 1 = 1 mark)

- 2 Magnesium reacts with hydrochloric acid.



Which statement about this reaction is correct?

- A magnesium atoms act as oxidising agents
- B hydrogen molecules act as reducing agents
- C hydrogen ions act as oxidising agents
- D chloride ions act as oxidising agents

(Total for Question 2 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

- 3 Barium chloride solution, $\text{BaCl}_2(\text{aq})$, reacts with gallium sulfate solution, $\text{Ga}_2(\text{SO}_4)_3(\text{aq})$, to form a precipitate of barium sulfate, $\text{BaSO}_4(\text{s})$.

What is the **minimum** volume of $0.100 \text{ mol dm}^{-3}$ barium chloride needed to precipitate all the sulfate ions in 200 cm^3 of 0.05 mol dm^{-3} gallium sulfate?

- A 100 cm^3
- B 200 cm^3
- C 300 cm^3
- D 400 cm^3

(Total for Question 3 = 1 mark)

- 4 Magnesium nitrate decomposes on heating.



What is the total volume of gas formed at room temperature and pressure (r.t.p.) when 0.005 mol of magnesium nitrate decomposes completely?

[Data: Molar volume of a gas at r.t.p. = $24\,000 \text{ cm}^3 \text{ mol}^{-1}$]

- A 600 cm^3
- B 300 cm^3
- C 240 cm^3
- D 120 cm^3

(Total for Question 4 = 1 mark)

- 5 Which of these compounds does **not** produce a colour in a flame test, **and** produces an alkaline gas when warmed with sodium hydroxide solution?

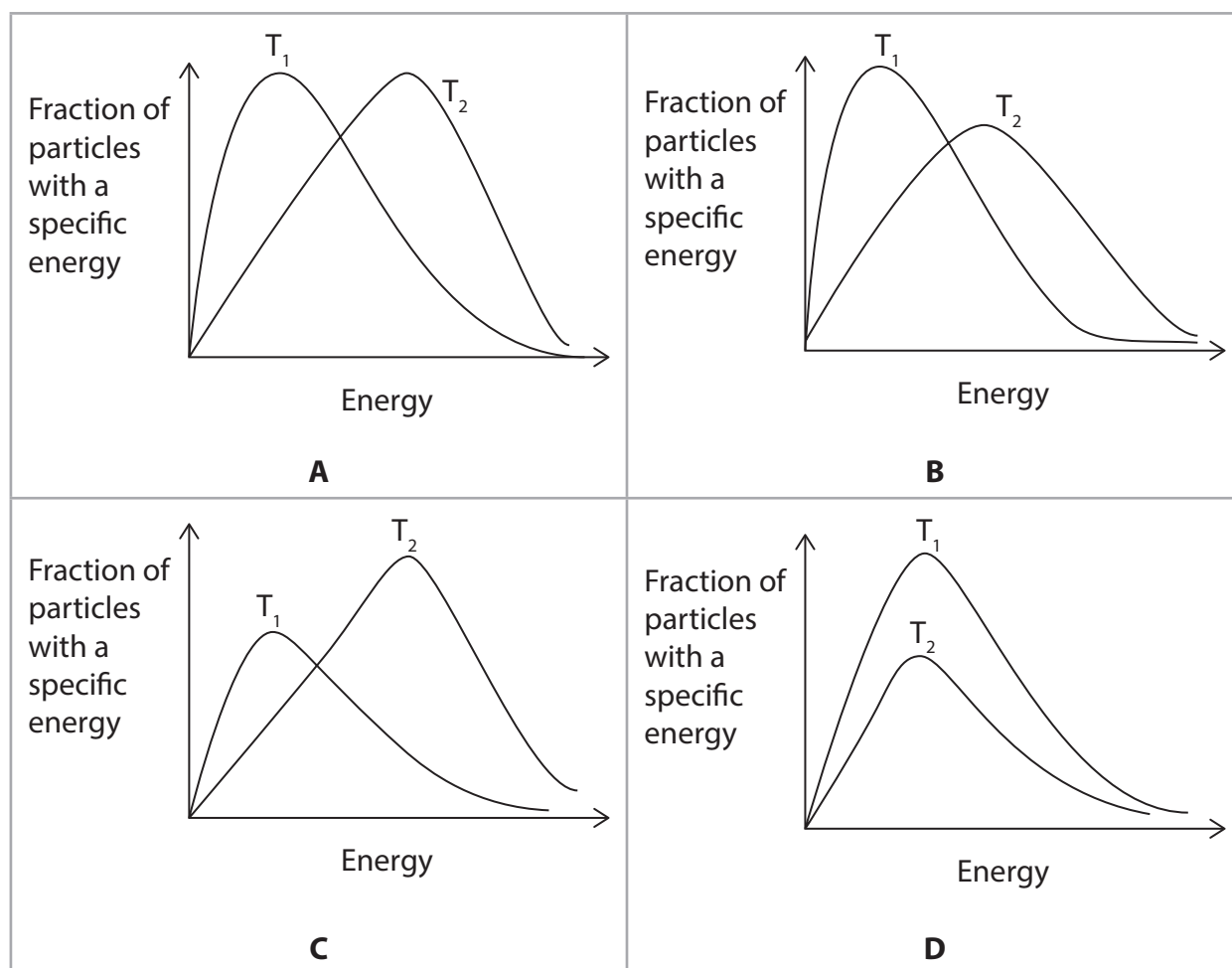
- A BeCl_2
- B $\text{Ca}(\text{OH})_2$
- C $\text{Mg}(\text{OH})_2$
- D NH_4Cl

(Total for Question 5 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

6 This question is about the Maxwell-Boltzmann energy distribution.

Which is the diagram for an increase in temperature from T_1 to T_2 ?



(Total for Question 6 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

- 7 The properties of Group 2 compounds change **down** the group from magnesium to barium.

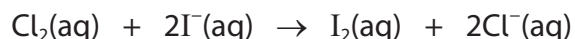
Which statement is correct?

- A thermal stability of Group 2 nitrates increases
- B thermal stability of Group 2 carbonates decreases
- C solubility of Group 2 sulfates increases
- D solubility of Group 2 hydroxides increases

(Total for Question 7 = 1 mark)

- 8 Chlorine is added to 2 cm³ of a dilute solution of potassium iodide.

The equation for the reaction between chlorine and iodide ions is shown.



(a) Which statement is correct?

- A iodide ions oxidise chlorine
- B iodide ions reduce chlorine
- C chlorine reduces iodide ions
- D chlorine is neither oxidised nor reduced

- (b) When the reaction is complete, 10 cm³ of cyclohexane (density = 0.79 g cm⁻³) is added. The mixture is shaken and left to settle into two layers.

Which description of one of these layers is correct?

- A the upper layer is purple
- B the lower layer is purple
- C the upper layer is brown
- D the lower layer is brown

(Total for Question 8 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

9 How many moles are there in 15.1 cm³ of liquid propan-1-ol?

[Data: Density of propan-1-ol = 0.80 g cm⁻³ M_r of propana-1-ol = 60]

- A** $(0.80 \times 15.1) \div 60$
- B** $0.80 \div (60 \times 15.1)$
- C** $60 \div (0.80 \times 15.1)$
- D** $(60 \times 15.1) \div 0.80$

(Total for Question 9 = 1 mark)

10 The oxidation of propan-1-ol by acidified potassium dichromate(VI) forms propanoic acid with a yield of 36% by mass.

What mass of propan-1-ol is needed to form 37.0 g of propanoic acid in this reaction?

[Data: M_r of propanoic acid = 74 M_r of propan-1-ol = 60]

- A** 10.8 g
- B** 36.0 g
- C** 40.8 g
- D** 83.3 g

(Total for Question 10 = 1 mark)

11 Which equation represents the standard enthalpy change of formation, $\Delta_f H^\ominus$, of hydrogen iodide?

- A** $\text{H(g)} + \text{I(g)} \rightarrow \text{HI(g)}$
- B** $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightarrow 2\text{HI(g)}$
- C** $\frac{1}{2}\text{H}_2\text{(g)} + \frac{1}{2}\text{I}_2\text{(g)} \rightarrow \text{HI(g)}$
- D** $\frac{1}{2}\text{H}_2\text{(g)} + \frac{1}{2}\text{I}_2\text{(s)} \rightarrow \text{HI(g)}$

(Total for Question 11 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.



12 The equation for a reversible reaction is shown.



What effect will each change have on the rate of reaction and the equilibrium yield of phosphorus(V) chloride?

(a) Increasing the temperature at constant pressure.

	Effect on rate of reaction	Effect on yield of $\text{PCl}_5(\text{g})$
A	increase	decrease
B	decrease	decrease
C	increase	increase
D	decrease	increase

(b) Increasing the pressure at constant temperature.

	Effect on rate of reaction	Effect on yield of $\text{PCl}_5(\text{g})$
A	increase	decrease
B	decrease	decrease
C	increase	increase
D	decrease	increase

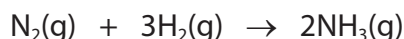
(Total for Question 12 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

13 Consider the bond enthalpy values given in the table.

Bond	Bond enthalpy / kJ mol^{-1}
$\text{N}\equiv\text{N}$	+945
$\text{H}-\text{H}$	+436
$\text{N}-\text{H}$	+391

The equation for the reaction between nitrogen and hydrogen is shown.



What is the enthalpy change, in kJ mol^{-1} , for this reaction?

- A** +1080
- B** +852
- C** -93
- D** -529

(Total for Question 13 = 1 mark)

14 The enthalpy change of combustion of ethanol was determined by using a spirit burner containing ethanol to heat 250 cm^3 of water in a copper calorimeter. The experimental value obtained was less exothermic than the Data Booklet value.

Which is the **least** likely reason for this difference in the enthalpy values?

- A** heat loss from the copper calorimeter
- B** incomplete combustion of the ethanol
- C** loss of ethanol by evaporation
- D** use of non-standard conditions

(Total for Question 14 = 1 mark)

15 What is the most significant factor determining the trend in the rate of hydrolysis of halogenoalkanes?

- A** the electronegativity of the halogen
- B** the magnitude of the halogen ionisation energy
- C** the oxidising ability of the halogen
- D** the carbon-halogen bond strength

(Total for Question 15 = 1 mark)

16 Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$, and propanone, CH_3COCH_3 , are isomers.

(a) Which m/z peak would **not** be expected in the mass spectrum of propanone?

(1)

- A** 15
- B** 29
- C** 43
- D** 58

(b) Propanal and propanone can be distinguished by chemical tests.

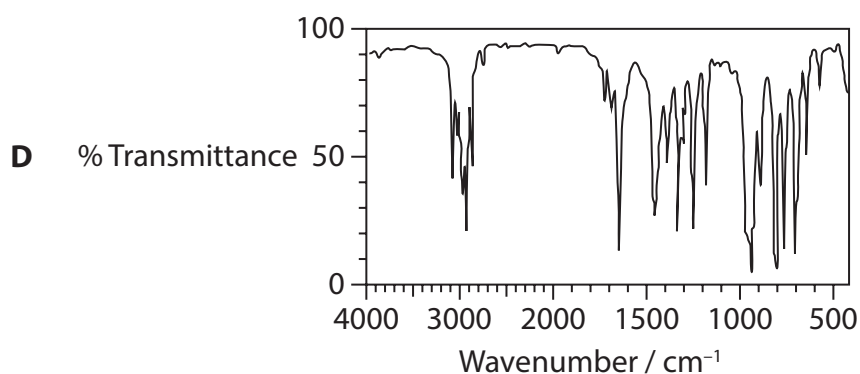
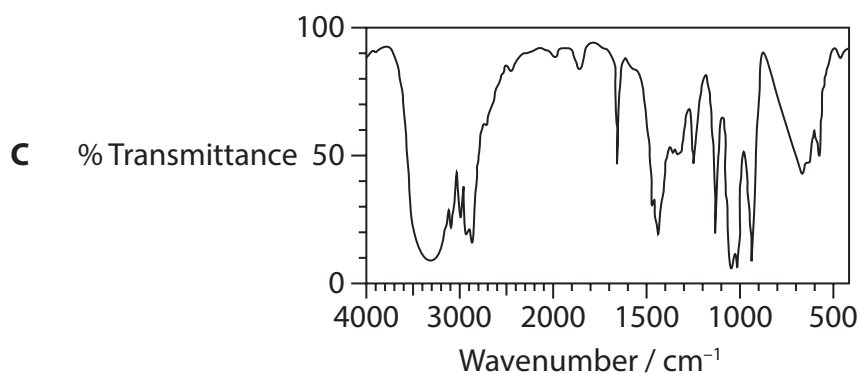
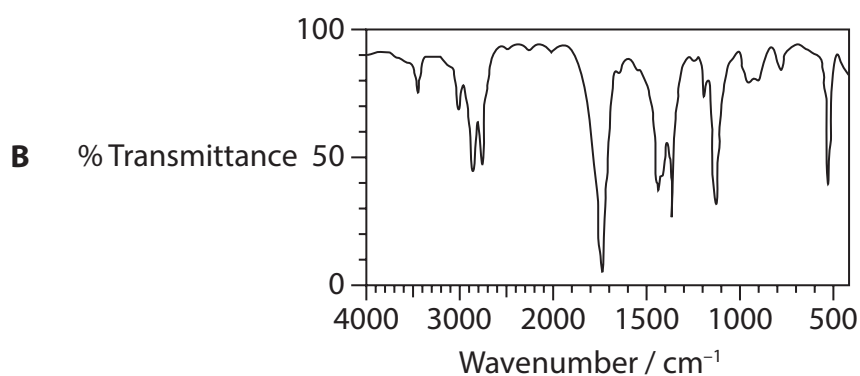
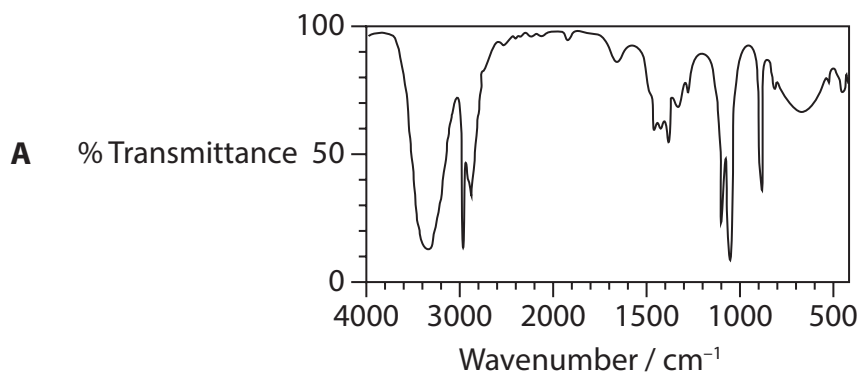
Which pair of observations is correct?

(1)

	Test	Observation with propanal	Observation with propanone
A	warm with Fehling's solution	no change	red precipitate
B	add solid phosphorus(V) chloride	no change	misty fumes
C	warm with acidified potassium dichromate(VI)	turns green	no change
D	add sodium hydrogencarbonate	fizzes	no change

(Total for Question 16 = 2 marks)

17 Which could be the infrared spectrum of $\text{CH}_2=\text{CHCH}_2\text{OH}$?



(Total for Question 17 = 1 mark)

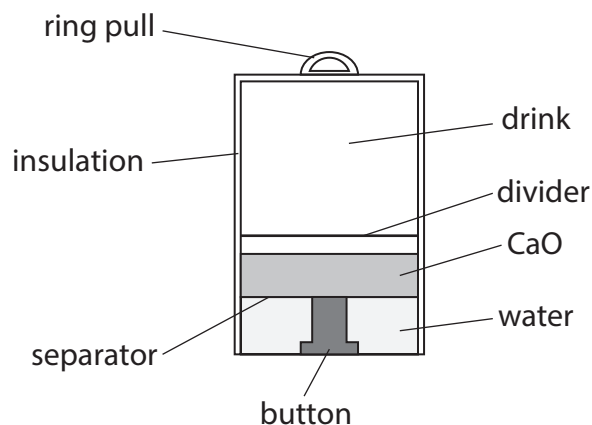
TOTAL FOR SECTION A = 20 MARKS

SECTION B

Answer ALL the questions in the Answer Booklet.

18 Salts have many uses.

- (a) The reaction between calcium oxide and water to produce calcium hydroxide solution is exothermic, and is used to warm canned drinks. A typical arrangement is shown.



The enthalpy change of reaction is $-65.1 \text{ kJ mol}^{-1}$.

- (i) Calculate the mass of calcium oxide that would be required to raise the temperature of 200 cm^3 of drink by 40°C .

[Data: Assume the specific heat capacity of the solution = $4.18 \text{ J g}^{-1}^\circ\text{C}^{-1}$
density of solution = 1.00 g cm^{-3}]

(3)

- (ii) Suggest **two** reasons why the can is insulated.

(2)

- (b) Ammonium ethanoate, $\text{CH}_3\text{COONH}_4(\text{s})$, is used to control the pH of foods. It can be formed by reaction of pure ethanoic acid, $\text{CH}_3\text{COOH}(\text{l})$, with ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3(\text{s})$.

Calculate the standard enthalpy change for this reaction by completing the Hess cycle in the Answer Book **and** using the data shown.

(5)

Compound	Enthalpy change of formation / kJ mol^{-1}
$\text{CH}_3\text{COOH}(\text{l})$	-484.5
$(\text{NH}_4)_2\text{CO}_3(\text{s})$	-939.9
$\text{CH}_3\text{COONH}_4(\text{s})$	-586.3
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O}(\text{l})$	-285.8

- (c) Ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is an ingredient in cleaning solutions for camera lenses.

These are aqueous solutions which contain no more than 3.60 g of ammonium carbonate in 200 cm^3 of solution.

Calculate the maximum concentration, in mol dm^{-3} , of ammonium carbonate in such a solution.

(2)

(Total for Question 18 = 12 marks)

19 This is a question about alcohols.

- (a) There are two alcohol structural isomers with the molecular formula, C_3H_8O .

Complete the table with the **skeletal** formula of these isomers, their IUPAC names and the classification of the type of alcohol in each case.

(3)

Skeletal formula	Name	Classification

- (b) Ethanol can be oxidised by acidified sodium dichromate(VI) to ethanal and then to ethanoic acid. The apparatus may be set up in two ways.

Reflux apparatus	Distillation apparatus
Diagram of a reflux apparatus. It consists of a pear-shaped flask containing anti-bumping granules at the bottom. The flask is heated from below. A vertical Liebig condenser is attached to the neck of the flask. Water enters the condenser from the bottom and exits from the top. The top of the condenser is open to the atmosphere.	Diagram of a distillation apparatus. It consists of a round-bottomed flask containing anti-bumping granules at the bottom. The flask is heated from below. A thermometer is inserted into the neck of the flask. A Liebig condenser is attached to the side of the flask, angled downwards. Water enters the condenser from the bottom and exits from the top. The end of the condenser is positioned over a beaker to collect the distillate.

- (i) Complete the ionic half-equation for the reduction of the dichromate(VI) ions to chromium(III) ions.

State symbols are not required.

(2)



- (ii) Describe how the reflux apparatus ensures that any ethanal initially produced is further oxidised to ethanoic acid.

(1)

- (iii) The distillation apparatus effectively separates ethanal from ethanol because of the large difference in boiling temperatures, which is a result of the hydrogen bonding between the molecules in ethanol.

Compound	Boiling temperature / °C
Ethanol, CH ₃ CH ₂ OH	79
Ethanal, CH ₃ CHO	21

Draw a labelled diagram showing a hydrogen bond between two ethanol molecules.

Clearly indicate any relevant dipoles, lone pairs of electrons and relevant bond angles involved.

(3)

- (iv) State the reason why hydrogen bonds do **not** form between ethanal molecules.

(1)

- (c) Alcohols can be converted into halogenoalkanes.

- (i) Write the equation for the reaction between methanol, CH₃OH, and phosphorus(V) chloride, PCl₅.

(1)

- (ii) State the observation that would be made from this reaction.

(1)

(d) Alcohols can be produced from the reaction of halogenoalkanes with aqueous alkali.

- (i) Draw the mechanism for this reaction with 1-bromopropane. Include curly arrows, and relevant dipoles and lone pairs.

(3)

- (ii) The reaction of 1-bromopropane with concentrated alcoholic alkali forms a different organic product.

Give the name of this type of reaction and the **displayed formula** of the organic product.

(2)

(Total for Question 19 = 17 marks)

20 This question is about trends in the Periodic Table.

*(a) The boiling temperatures of some isoelectronic hydrides are shown.

Hydride	CH ₄	NH ₃	H ₂ O	HF
Boiling temperature / °C	−161	−33	100	20

Explain the differences in these boiling temperatures by considering all the intermolecular forces involved.

Detailed descriptions of the intermolecular forces involved are not required.

(6)

(b) Compare and contrast the reactions of concentrated sulfuric acid with solid sodium chloride and with solid sodium bromide.

(4)

(Total for Question 20 = 10 marks)

TOTAL FOR SECTION B = 39 MARKS

SECTION C

Answer ALL the questions. Write your answers in the spaces provided in the Answer Booklet.

21 Major volcanic eruptions, such as the one in Indonesia in 1815, eject large amounts of ash and gases into the atmosphere. The gases include carbon dioxide, water vapour, hydrogen chloride and sulfur dioxide.

- (a) (i) Complete the dot-and-cross diagram to show a possible arrangement of the outer shell electrons in a molecule of sulfur dioxide.
Use dots (•) for the sulfur electrons and crosses (×) for the oxygen electrons.

(2)

- (ii) Suggest a value for the bond angle.

(1)

- (b) The levels of sulfur dioxide in the atmosphere can be measured by collection and titration with sodium hydroxide solution.

One possible method is:

- air containing sulfur dioxide is bubbled through hydrogen peroxide solution
- all the sulfur dioxide reacts to form 40.0 cm³ of dilute sulfuric acid, solution **Z**



- a pipette is used to remove 10.0 cm³ portions of solution **Z**
- each portion is titrated with 0.00400 mol dm⁻³ sodium hydroxide

The results are shown in the table.

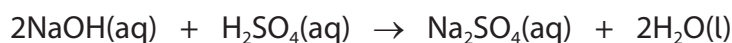
Titration	1	2	3
Final volume / cm ³	22.55	44.70	22.05
Initial volume / cm ³	0.00	22.55	0.00
Titre / cm ³	22.55	22.15	22.05

The mean titre is 22.10 cm^3 .

- (i) Give a reason why a further titration was not carried out.

(1)

- (ii) Calculate the number of moles of sulfuric acid in the 40.0 cm^3 of solution **Z**.



(3)

- (iii) Deduce the number of moles of sulfur dioxide bubbled through the hydrogen peroxide solution, using your answer from (b)(ii) and the equation shown.



(1)

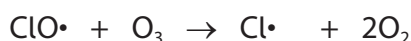
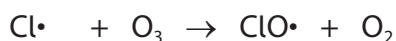
- (iv) The air containing sulfur dioxide was bubbled through the hydrogen peroxide solution at a rate of $15\text{ dm}^3\text{ min}^{-1}$ for 20 minutes.

Calculate the concentration of sulfur dioxide in the air in parts per million (ppm) by volume.

[Data: Molar volume of a gas at room temperature and pressure = $24\text{ dm}^3\text{ mol}^{-1}$]

(3)

- (c) (i) During a volcanic eruption, hydrogen chloride gas is also released into the upper atmosphere. This produces some chlorine free radicals that react with ozone as shown.



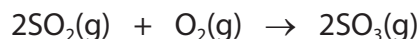
Deduce the overall equation for this reaction of ozone.
State symbols are not required.

(1)

- (ii) Give two reasons why the presence of a small number of chlorine free radicals in the upper atmosphere causes a large decrease in the amount of ozone.

(2)

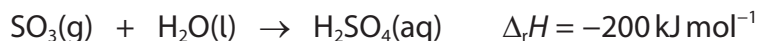
- (d) (i) Sulfur dioxide is converted into sulfur trioxide in the upper atmosphere in the presence of ultraviolet radiation.



Show, by use of all the relevant oxidation numbers, that this is a redox reaction.

(2)

- (ii) Sulfur trioxide reacts with water to form sulfuric acid.



Draw an enthalpy level diagram to show the enthalpy change for this reaction.

(2)

- (iii) Droplets of sulfuric acid are formed when sulfur trioxide reacts with water in the upper atmosphere. These droplets reflect ultraviolet radiation from the Sun back into space.

In the volcanic eruption in Indonesia in 1815, large amounts of both carbon dioxide and sulfur dioxide were released.

During the following year, there were global low temperatures.

Explain the relative effect of both gases on global temperatures after the volcanic eruption.

(3)

(Total for Question 21 = 21 marks)

TOTAL FOR SECTION C = 21 MARKS
TOTAL FOR PAPER = 80 MARKS

20

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Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International Advanced Level

Monday 13 October 2025

Morning (Time: 1 hour 30 minutes) **Paper reference** **WCH12/01A**

Chemistry
International Advanced Subsidiary/Advanced Level
UNIT 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols
Answer Booklet

You must have:
 Scientific calculator, Data Booklet, ruler and Question paper

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
 – *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
 – *use this as a guide as to how much time to spend on each question.*
- In the question marked with an **asterisk** (*), marks will be awarded for your ability to structure your answer logically, showing how the points that you make are related or follow on from each other where appropriate.
- A Periodic Table is printed on the back cover of this paper.

Advice

- Read each question carefully before you start to answer it.
- Show all your working in calculations and include units where appropriate.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A

Answer ALL the questions in this section.

You should aim to spend no more than 20 minutes on this section.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☒.

1

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 1 = 1 mark)

2

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 2 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA



3

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Total for Question 3 = 1 mark)**4**

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Total for Question 4 = 1 mark)**5**

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 5 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.



6

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 6 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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7

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 7 = 1 mark)

8 (a)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(b)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 8 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.



9

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 9 = 1 mark)

10

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 10 = 1 mark)

11

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 11 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.



12 (a)

☐ A

☐ B

☐ C

☐ D

(b)

☐ A

☐ B

☐ C

☐ D

(Total for Question 12 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.



13

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 13 = 1 mark)

14

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 14 = 1 mark)



15☐ **A**☐ **B**☐ **C**☐ **D****(Total for Question 15 = 1 mark)****16 (a)****(1)**☐ **A**☐ **B**☐ **C**☐ **D****(b)****(1)**☐ **A**☐ **B**☐ **C**☐ **D****(Total for Question 16 = 2 marks)**

17

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 17 = 1 mark)

TOTAL FOR SECTION A = 20 MARKS



SECTION B

Answer ALL the questions. Write your answers in the spaces provided.

18 (a) (i)

(3)

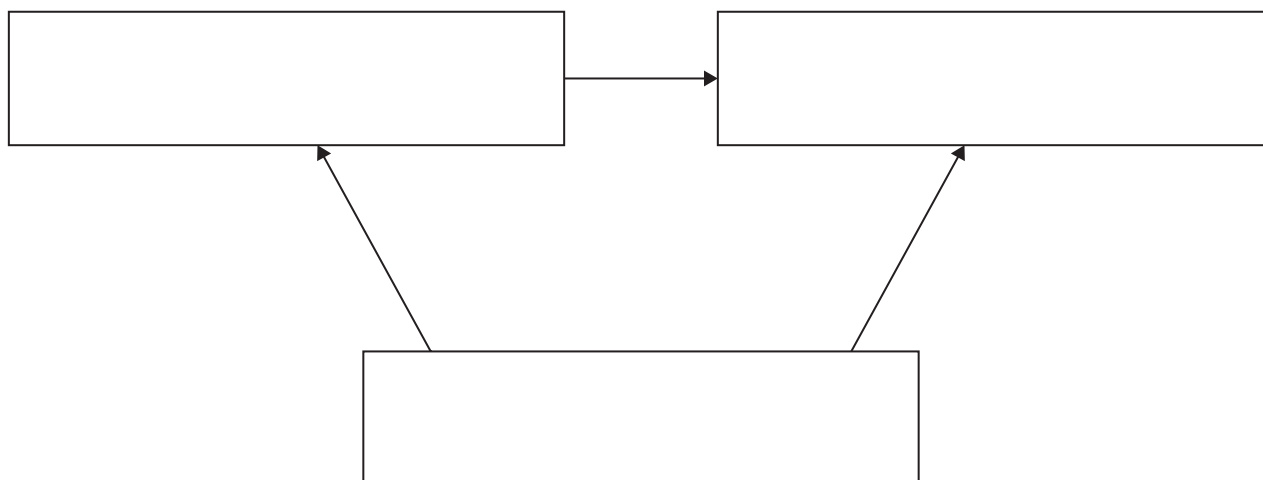
(ii)

(2)



(b)

(5)



(c)

(2)

(Total for Question 18 = 12 marks)

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19 (a)

(3)

Skeletal formula	Name	Classification



(b) (i)

(2)



(ii)

(1)

(iii)

(3)



(iv)

(1)

(c)

(i)

(1)

(ii)

(1)

(d) (i)

(3)

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DO NOT WRITE IN THIS AREA



(ii)

(2)

Name of type of reaction

.....

(Total for Question 19 = 17 marks)

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20

*(a)

(6)

Handwriting practice area with 20 horizontal dotted lines.



Question 20(a) continued

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DO NOT WRITE IN THIS AREA

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(b)

(4)

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(Total for Question 20 = 10 marks)

TOTAL FOR SECTION B = 39 MARKS



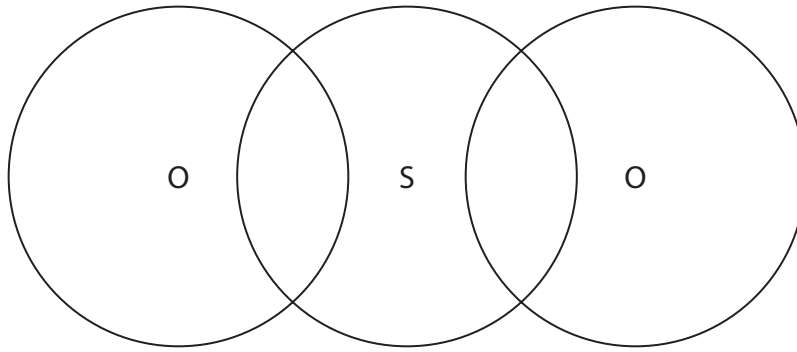
SECTION C

Answer ALL the questions. Write your answers in the spaces provided.

21

(a) (i)

(2)



(ii)

(1)

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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(b)

(i)

(1)

(ii)

(3)

(iii)

(1)



(iv)

(3)

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) (i)

(1)



(ii)

(2)

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(d) (i)

(2)

.....

.....

.....

.....

(ii)

(2)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(iii)

(3)

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(Total for Question 21 = 21 marks)

TOTAL FOR SECTION C = 21 MARKS
TOTAL FOR PAPER = 80 MARKS



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28

P 8 7 4 7 5 A 0 2 8 2 8

* Lanthanide series

* Actinide series

140	Ce	cerium	58	141	Pr	praseodymium	59	144	Nd	neodymium	60	[147]	Pm	promethium	61	150	Sm	samarium	62	152	Eu	europium	63	157	Gd	gadolinium	64	159	Tb	terbium	65	163	Dy	dysprosium	66	165	Ho	holmium	67	167	Er	erbium	68	169	Tm	thulium	69	173	Yb	ytterbium	70	175	Lu	lutetium	71
232	Th	thorium	90	[231]	Pa	protactinium	91	238	U	uranium	92	[237]	Np	neptunium	93	[242]	Pu	plutonium	94	[243]	Am	americium	95	[247]	Cm	curium	96	[245]	Bk	berkelium	97	[251]	Cf	californium	98	[254]	Es	einsteinium	99	[253]	Fm	fermium	100	[256]	Md	mercurium	101	[254]	No	nobelium	102	[257]	Lr	lawrencium	103

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